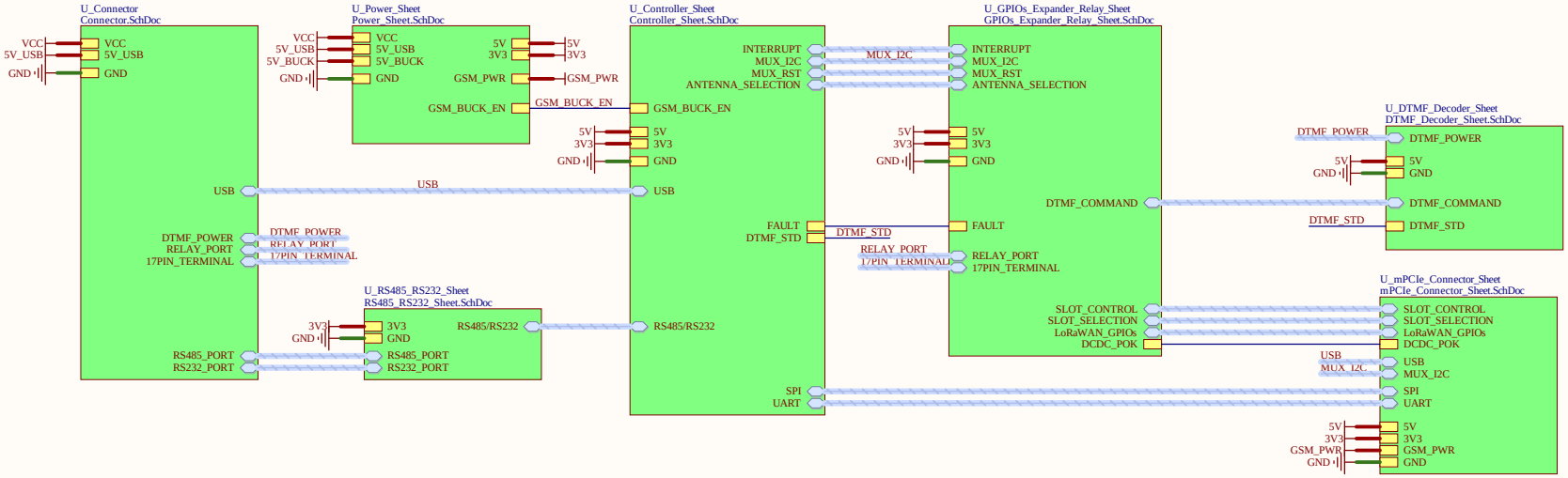
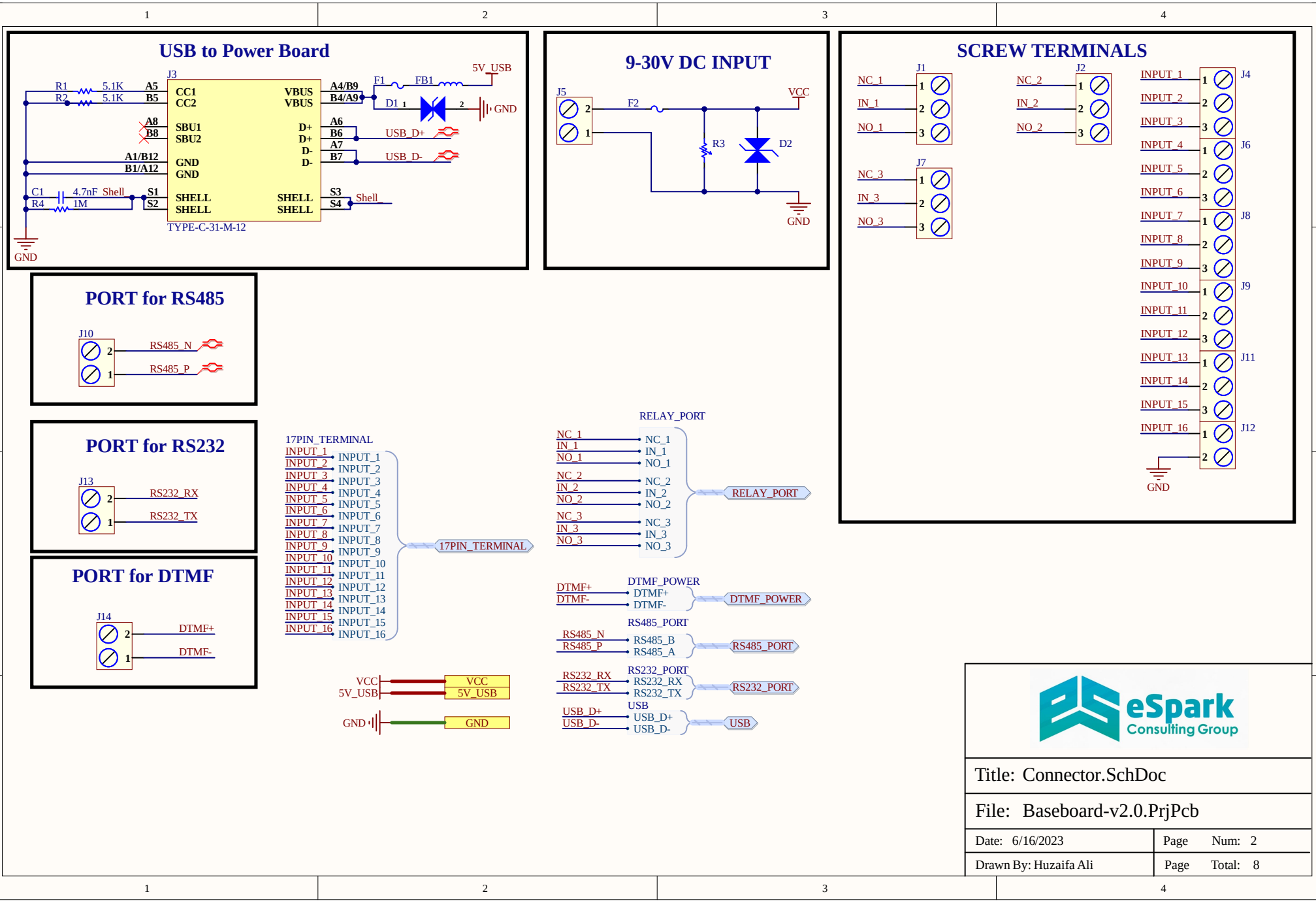
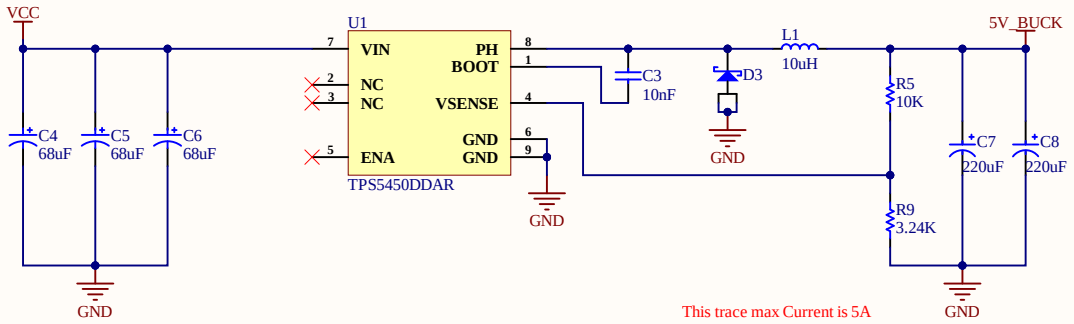


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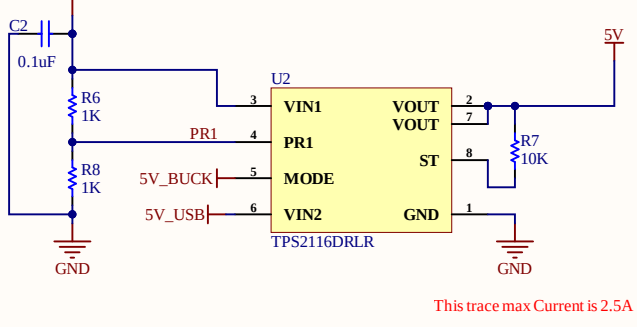
Title: Top_Sheet.SchDoc	
File: Baseboard-v2.0.PrjPcb	
Date: 6/16/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8


USB to Power Board
J3
R1 5.1K
R2 5.1K
A5
B5
CC1
CC2
A8
B8
SBU1
SBU2
A1/B12
B1/A12
GND
GND
C1 4.7nF Shell
R4 1M
S1
S2
SHELL
SHELL
VBUS
VBUS
D+
D+
D-
D-
SHELL
SHELL
A4/B9
B4/A9
A6
B6
A7
B7
S3
S4
F1 FB1
5V_USB
D1
GND
USB_D+
USB_D-
Shell_
TYPE-C-31-M-12
GND
9-30V DC INPUT
J5
F2
VCC
R3
D2
GND
SCREW TERMINALS
J1
NC_1
IN_1
NO_1
J2
NC_2
IN_2
NO_2
J7
NC_3
IN_3
NO_3
J4
INPUT_1
INPUT_2
INPUT_3
J6
INPUT_4
INPUT_5
INPUT_6
J8
INPUT_7
INPUT_8
INPUT_9
J9
INPUT_10
INPUT_11
INPUT_12
J11
INPUT_13
INPUT_14
INPUT_15
J12
INPUT_16
GND
PORT for RS485
J10
RS485_N
RS485_P
PORT for RS232
J13
RS232_RX
RS232_TX
PORT for DTMF
J14
DTMF+
DTMF-
17PIN_TERMINAL
INPUT_1
INPUT_2
INPUT_3
INPUT_4
INPUT_5
INPUT_6
INPUT_7
INPUT_8
INPUT_9
INPUT_10
INPUT_11
INPUT_12
INPUT_13
INPUT_14
INPUT_15
INPUT_16
17PIN_TERMINAL
RELAY_PORT
NC_1
IN_1
NO_1
NC_2
IN_2
NO_2
NC_3
IN_3
NO_3
RELAY_PORT
DTMF_POWER
DTMF+
DTMF-
DTMF_POWER
RS485_PORT
RS485_N
RS485_P
RS485_B
RS485_A
RS485_PORT
RS232_PORT
RS232_RX
RS232_TX
RS232_PORT
USB
USB_D+
USB_D-
USB
VCC
5V_USB
VCC
5V_USB
GND
GND
eSpark
Consulting Group
Title: Connector.SchDoc
File: Baseboard-v2.0.PrjPcb
Date: 6/16/2023
Page Num: 2
Drawn By: Huzaifa Ali
Page Total: 8

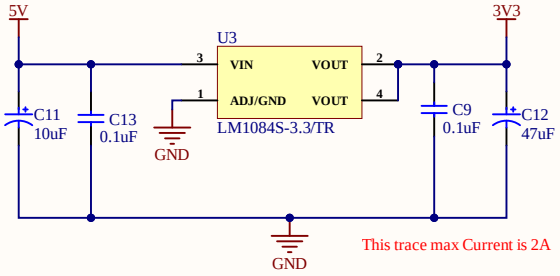
5V BUCK REGULATOR



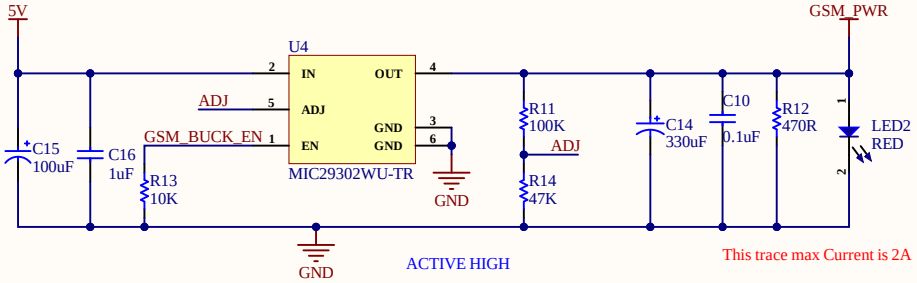
ORING CIRCUIT



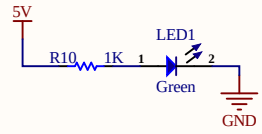
LDO 3.3V

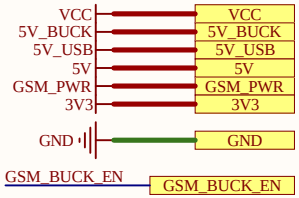


BUCK FOR GSM POWER

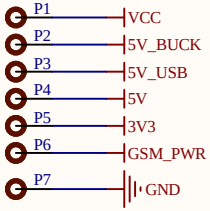


POWER LED



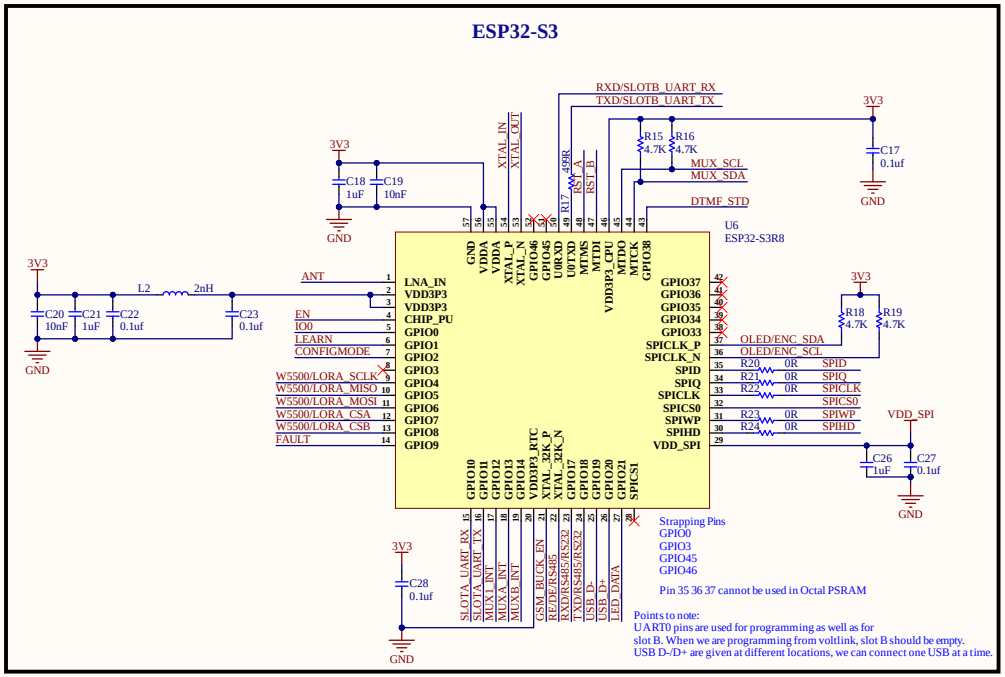


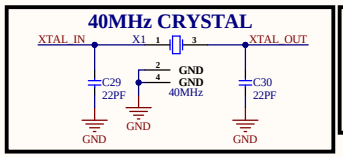
TEST POINTS

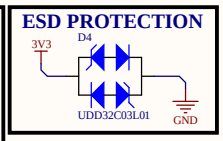


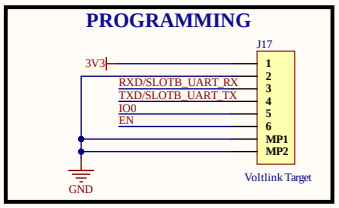
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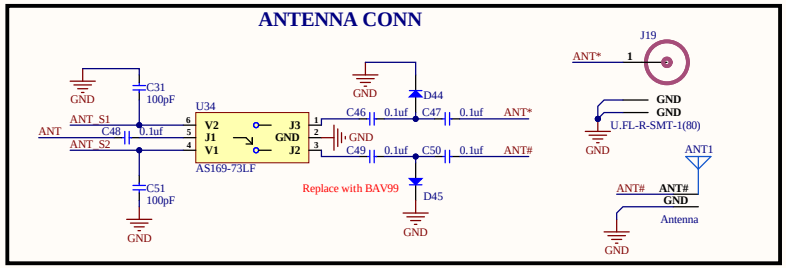
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File: Baseboard-v2.0.PrjPcb	
Date: 6/16/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 8

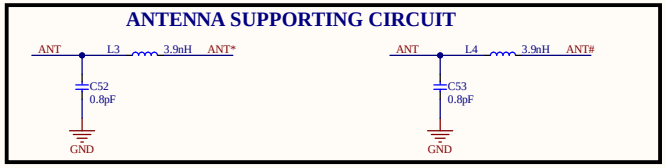


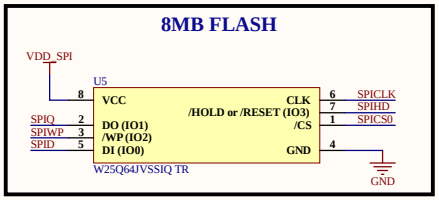


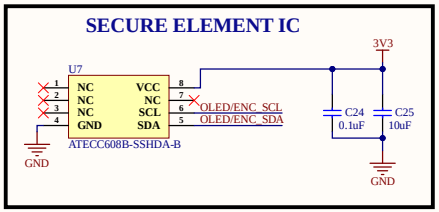


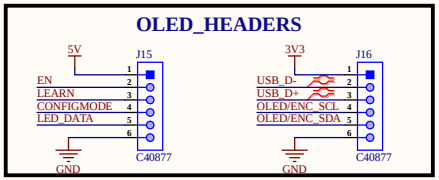


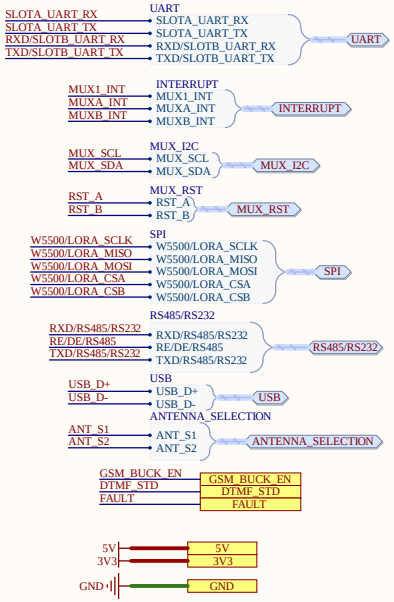












Notes for ESP32 Pinouts

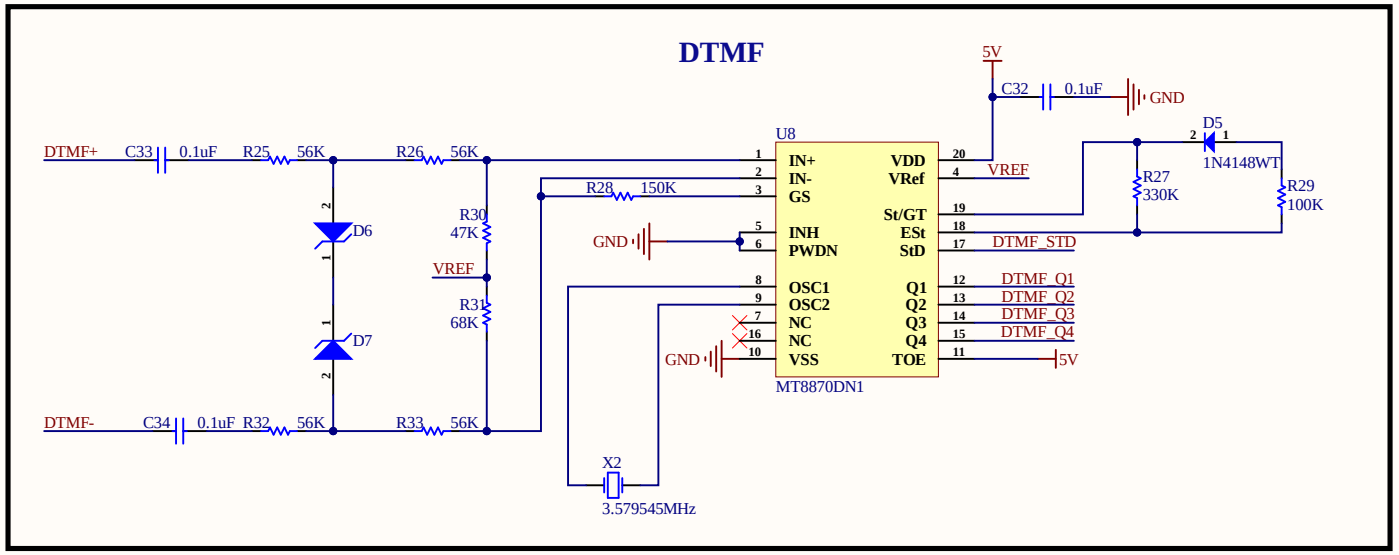
1 interrupt for 16 GPIOs
2 interrupts for Slot A/B

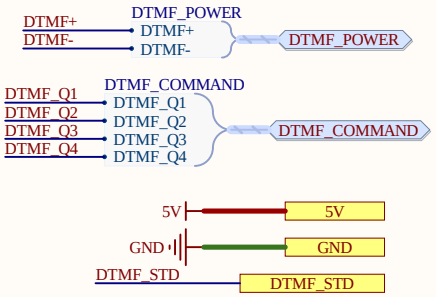
One I2C selected for OLED and Encryption IC
Second I2C selected for 4 MUX and given on mPCIE

Selection between antennas is given on MUX

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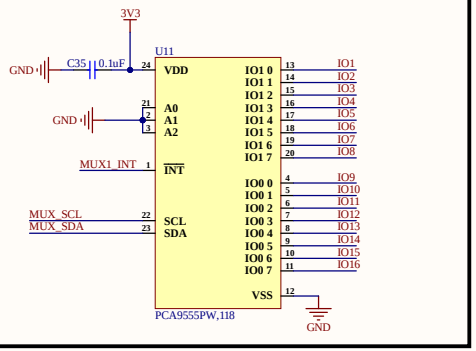
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Date: 6/16/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 8



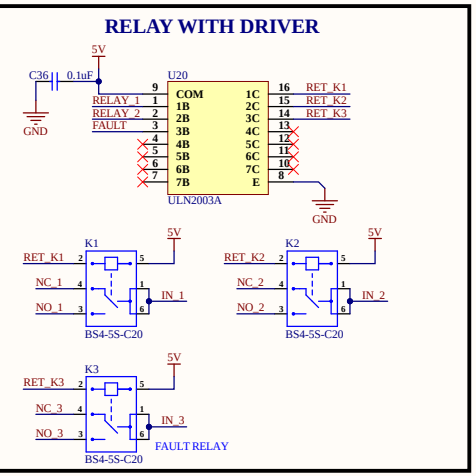


Title: DTMF_Decoder_Sheet.SchDoc	
File: Baseboard-v2.0.PrjPcb	
Date: 6/16/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8

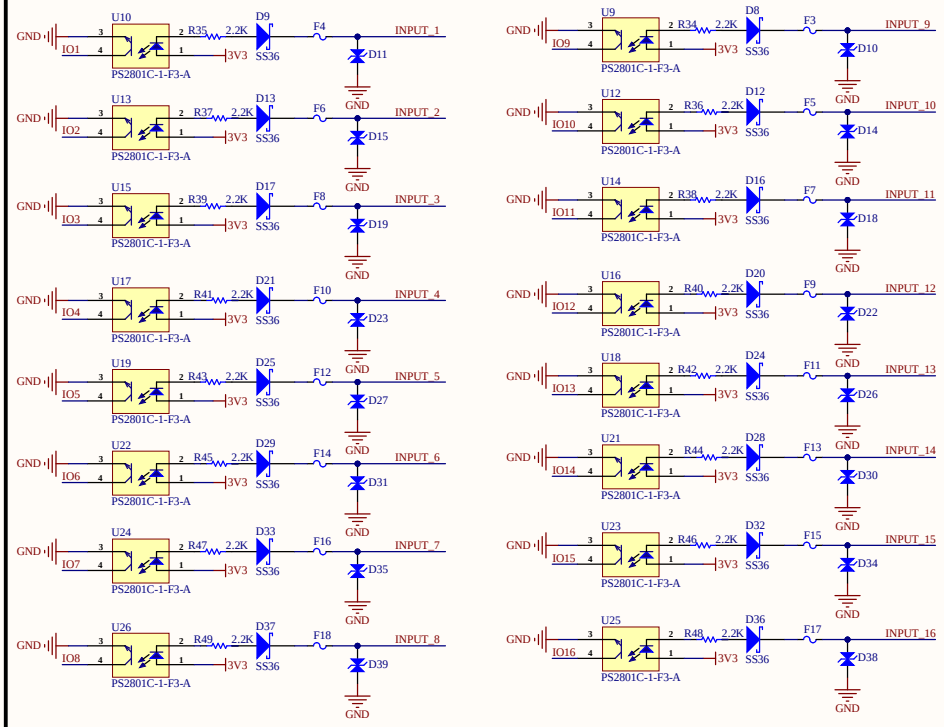
INPUT GPIO MULTIPLEXER

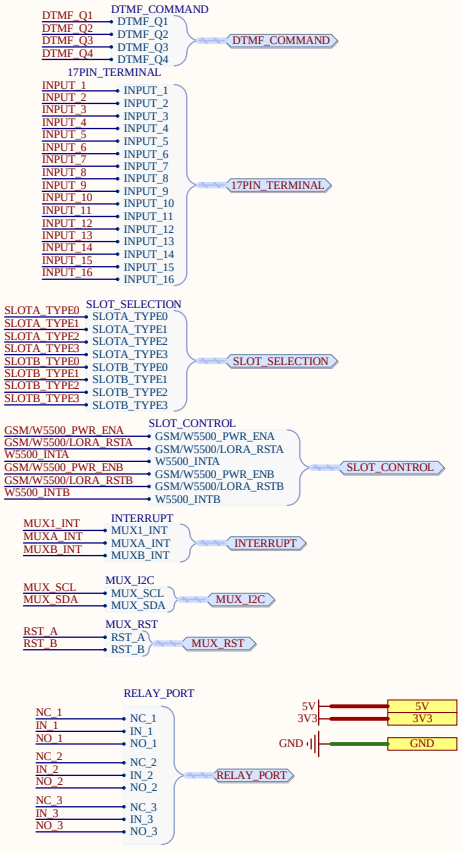


RELAY WITH DRIVER

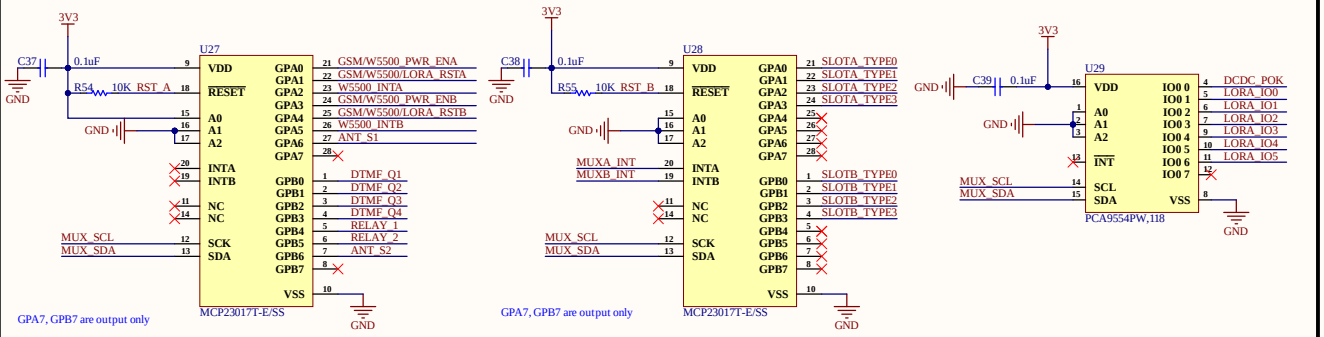


OPTO ISOLATION FOR GPIOs

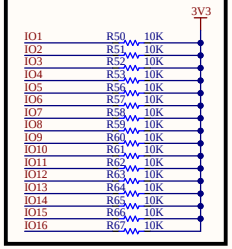




GPIO MULTIPLEXERS

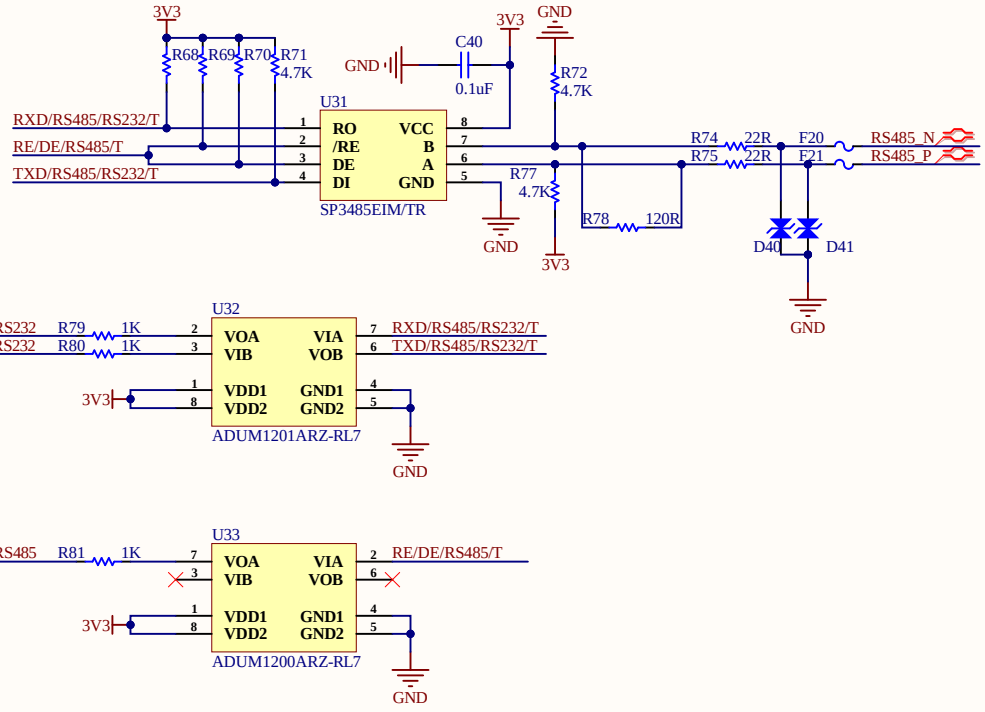


GPIO PULLUP

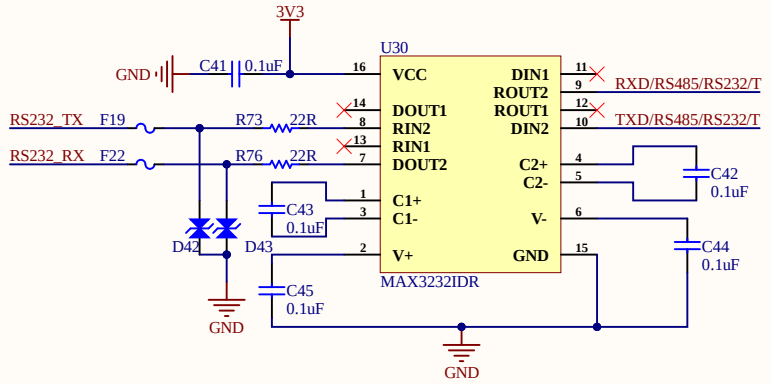


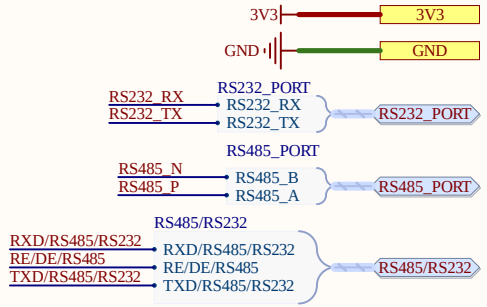
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File: Baseboard-v2.0.PrjPcb	
Date: 6/16/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 8

RS485 to UART



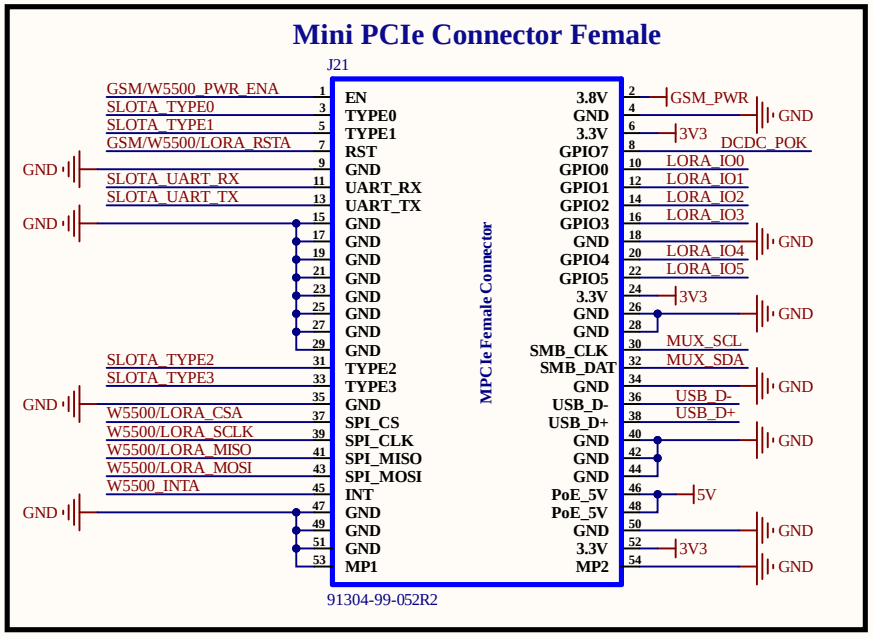
RS232 to UART

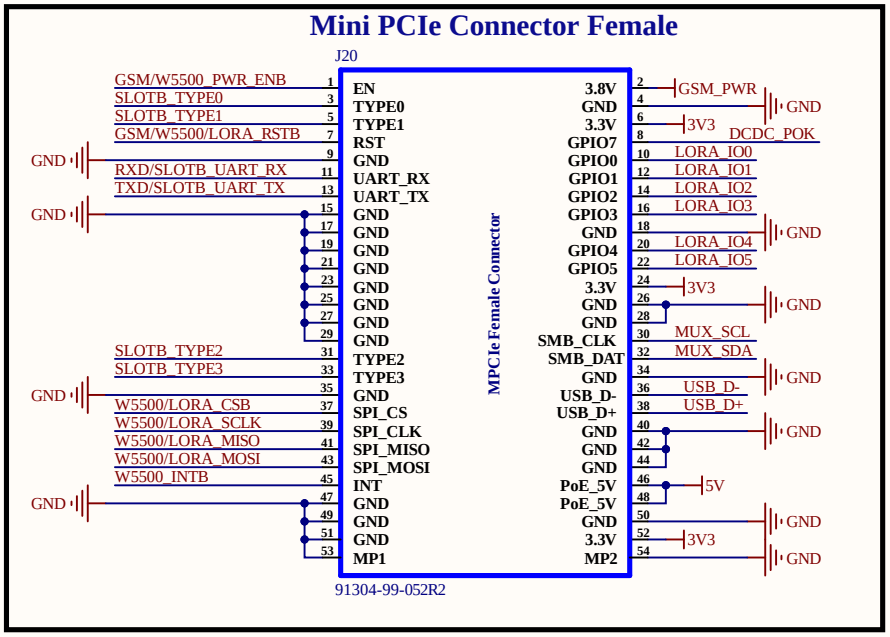


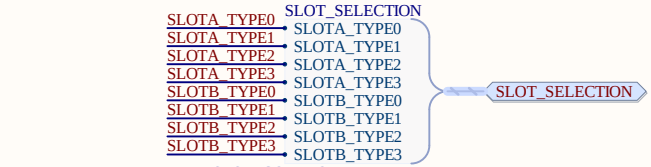


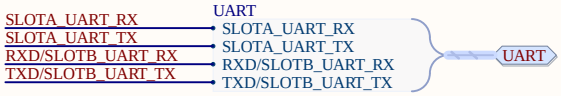
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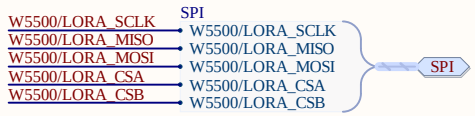
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Date: 6/16/2023	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 8

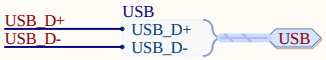




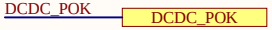


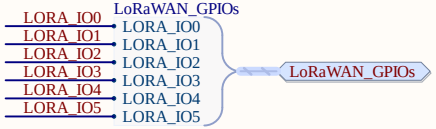


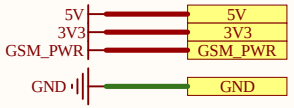












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Date: 6/16/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 8